城市NGP感知融合算法简介



深蓝学院公开课

2021.11.9

Dr. Feng Han



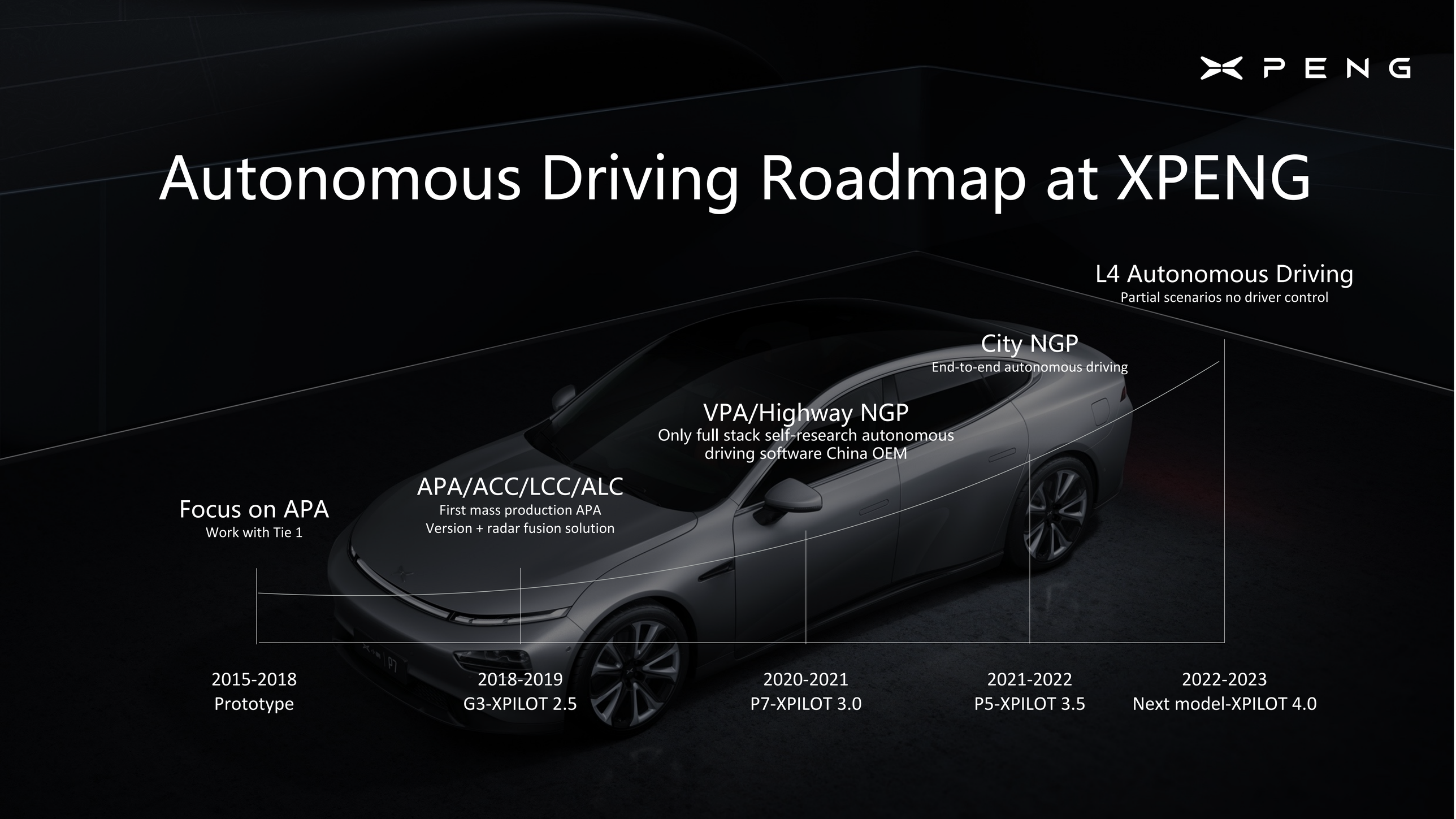
Principal Engineer/Director, Autonomous Driving

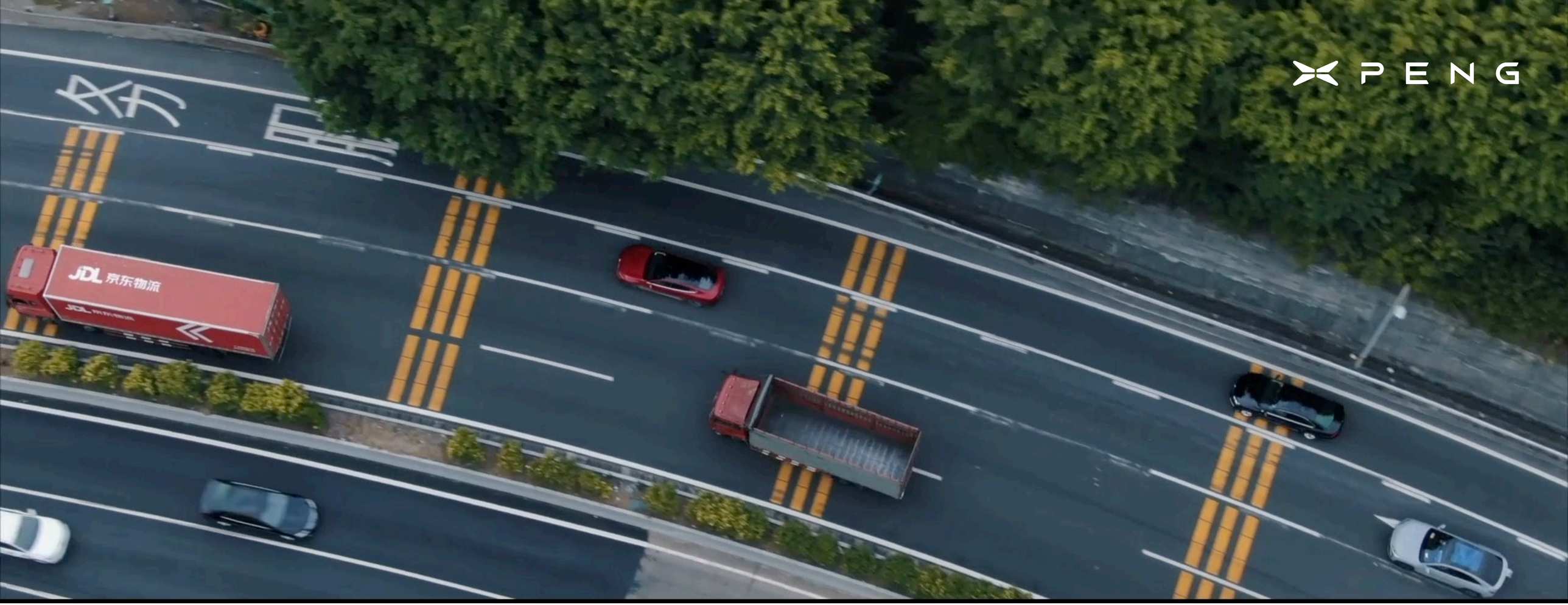
- Working on Advanced Autonomous Driving technology at Xpeng since Jan. 2019.
- Prior to joining XPENG, was with Corporate Research and Development, Qualcomm Inc., working on various key autonomous driving modules including sensor fusion, behavior prediction and behavior planning.
- Founding member of Origin Wireless, Inc, founded in 2013 in Maryland, pioneering commercialization of time reversal technology for wireless ambient sensing.

- Master Student Research on MIMO received Best Paper Award at IEEE WCNC2008.
- Ph.D. Research on time-reversal technology recognized by Invention of the Year Award at University of Maryland 2013.
- B.S. and M.S. in Electrical Engineering from Tsinghua University
- Ph.D. in Electrical Engineering, University of Maryland, 2013



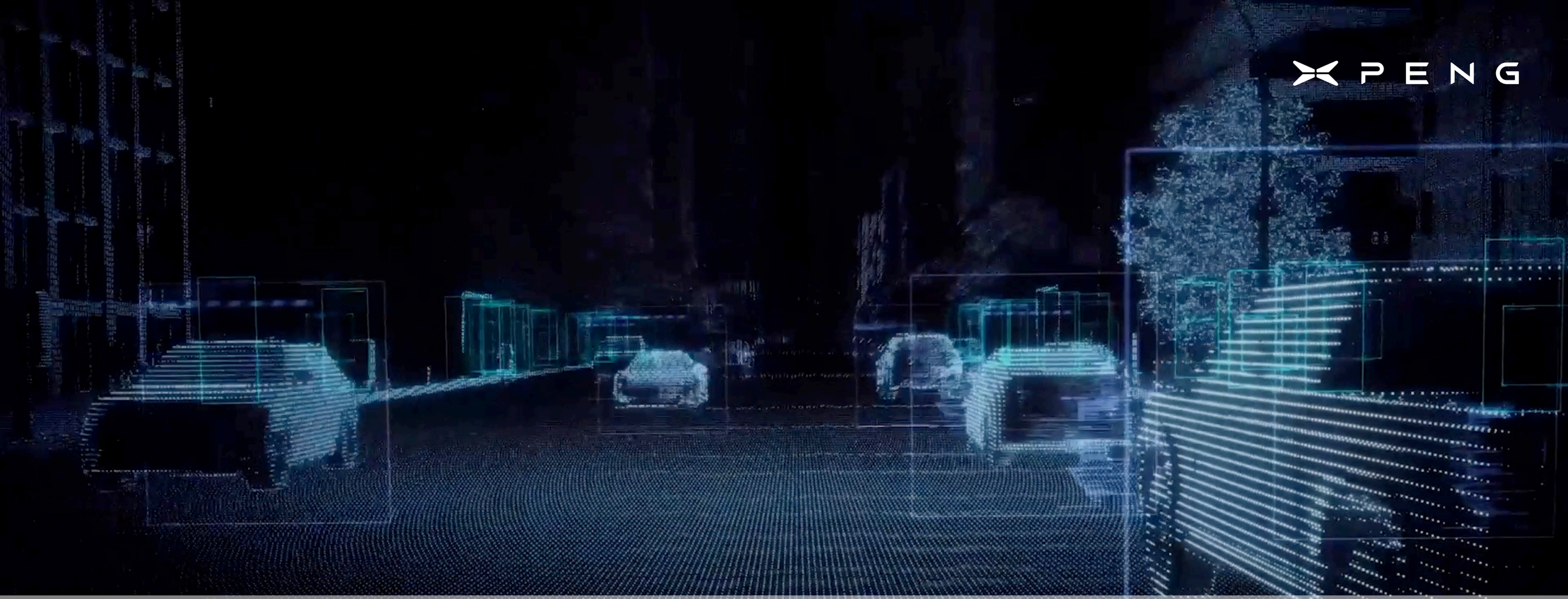
Autonomous Driving Roadmap at XPENG





NGP - Navigation Guided Pilot

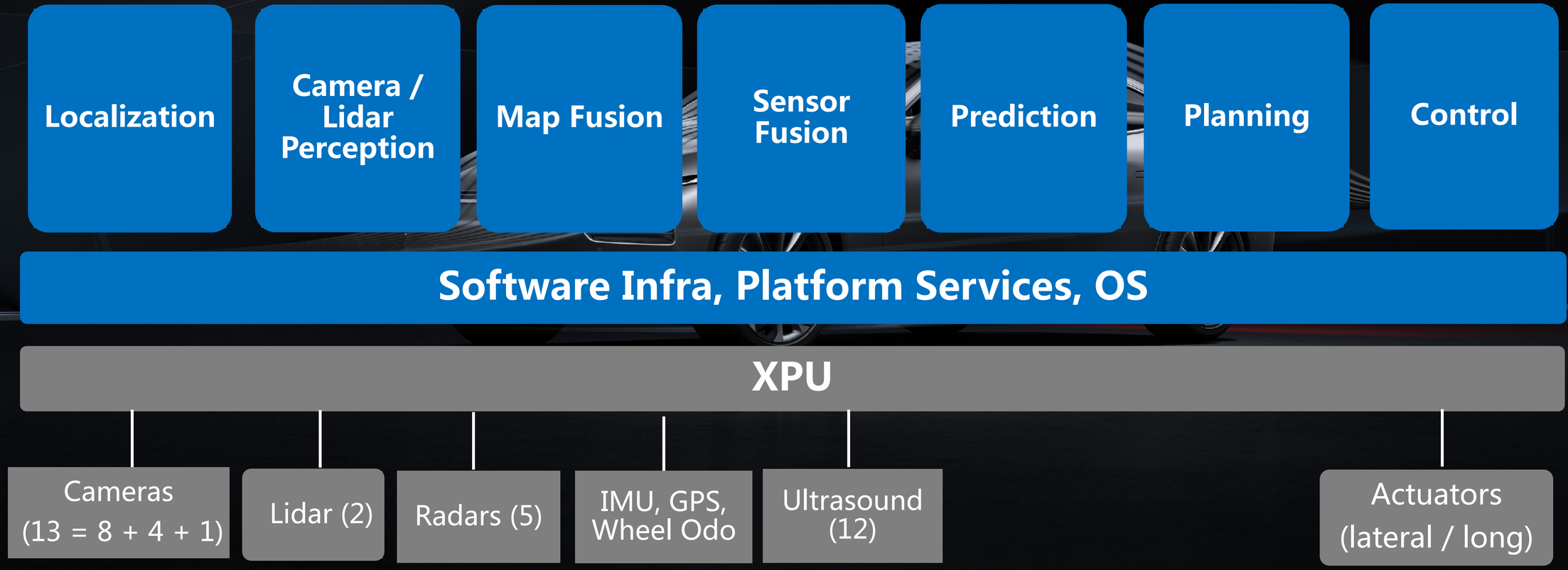
Based on the navigation route set by the user,
enable automatic navigation assistance driving from point A to point B.



CNGP City Navigation Guided Pilot

The first LiDAR-equipped mass production City NGP.
City NGP assists the driver to navigate the vehicle from point A to point B based on the navigation routes set by the driver in city driving scenarios

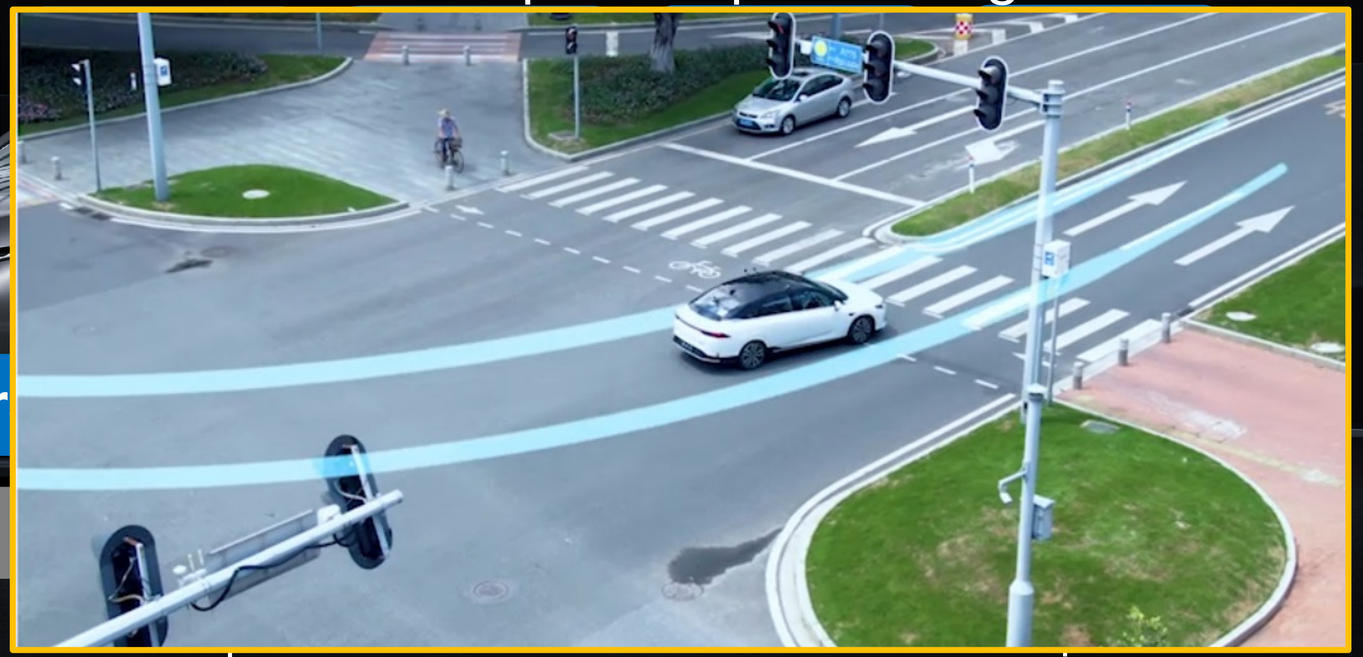
City NGP Technology Stack



City NGP assists the driver to navigate the vehicle from point A to point B in city driving environment (Intersections, Roundabouts, Traffic Lights, Pedestrians, etc…)

City NGP Technology Stack – Static Environment

Centimeter level precise positioning in HDMAP



Localization

Camera / Lidar Perception

Map Fusion (HDMAP)

Software Infr

Cameras (13 = 8 + 4 + 1)

Lidar (2)

Radars (5)

IMU, GPS, Wheel Odo

Ultrasound (12)

Actuators (lateral / long)

City NGP Technology Stack – Dynamic Environment

Localization

Camera / Lidar Perception

Map Fusion

Sensor Fusion

Software Infra, Platform

XPU

Cameras (13 = 8 + 4 + 1)

Lidar (2)

Radars (5)

IMU, GPS, Wheel Odo

Ultrasound (12)

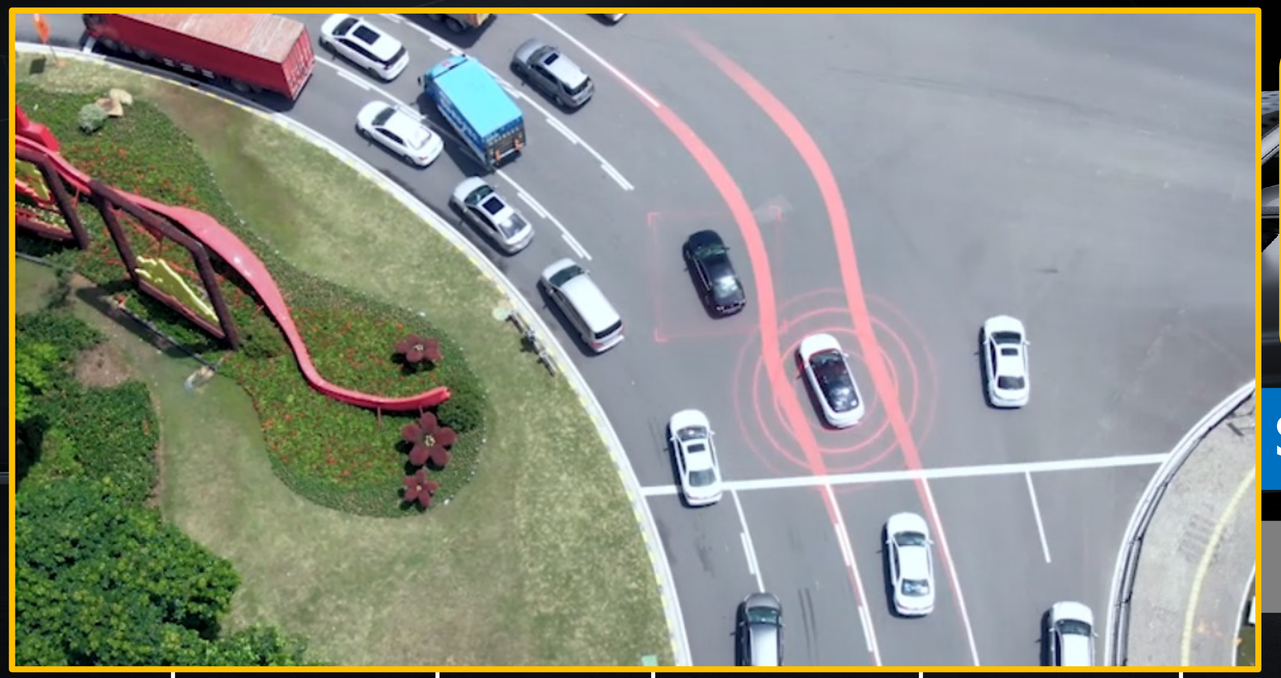
Centimeter level positioning for agents



Actuators (lateral / long)

City NGP Technology Stack – The Driver

Human like driving behavior utilizing precise prediction



Prediction

Planning

Control

Services, OS

Cameras (13 = 8 + 4 + 1)

Lidar (2)

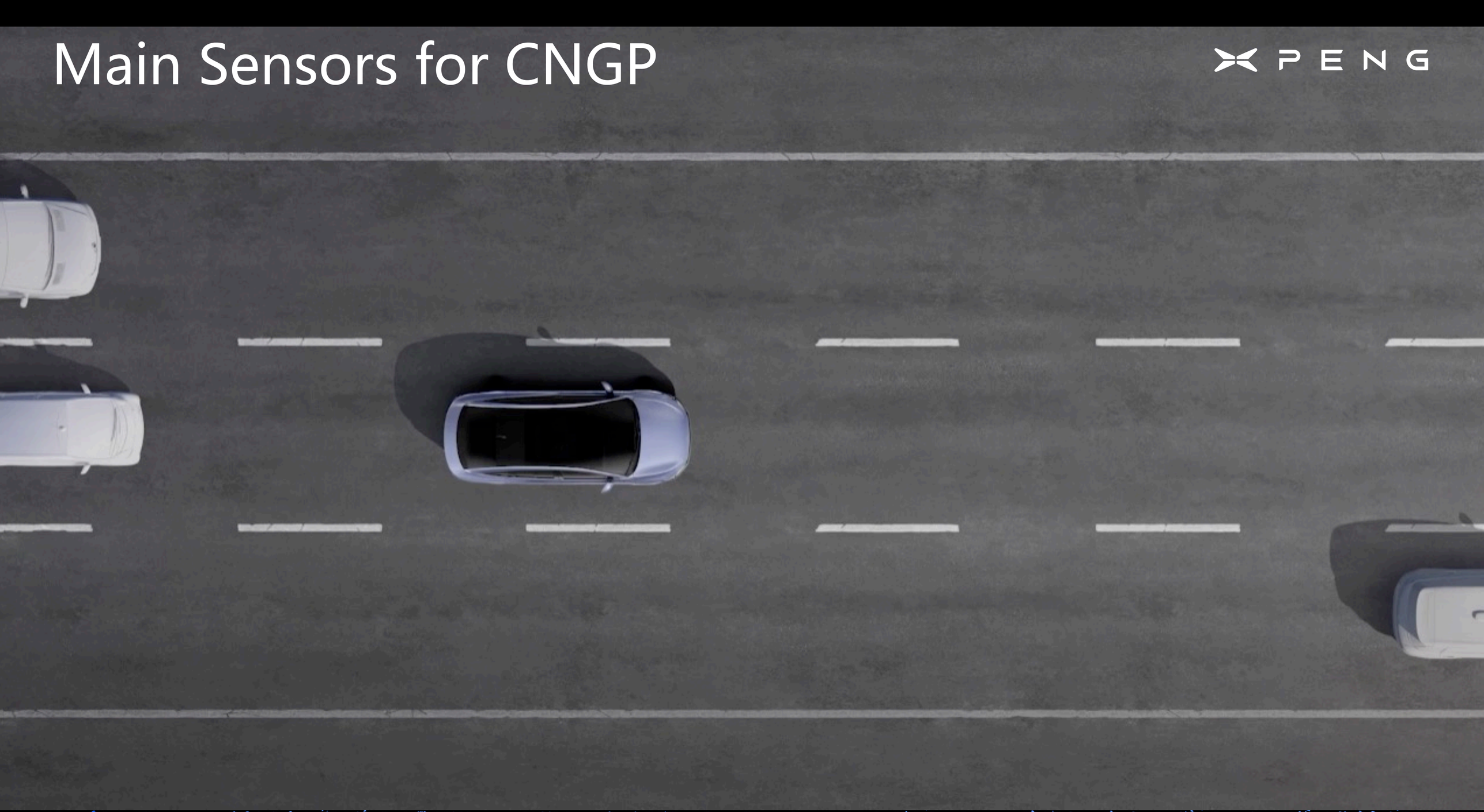
Radars (5)

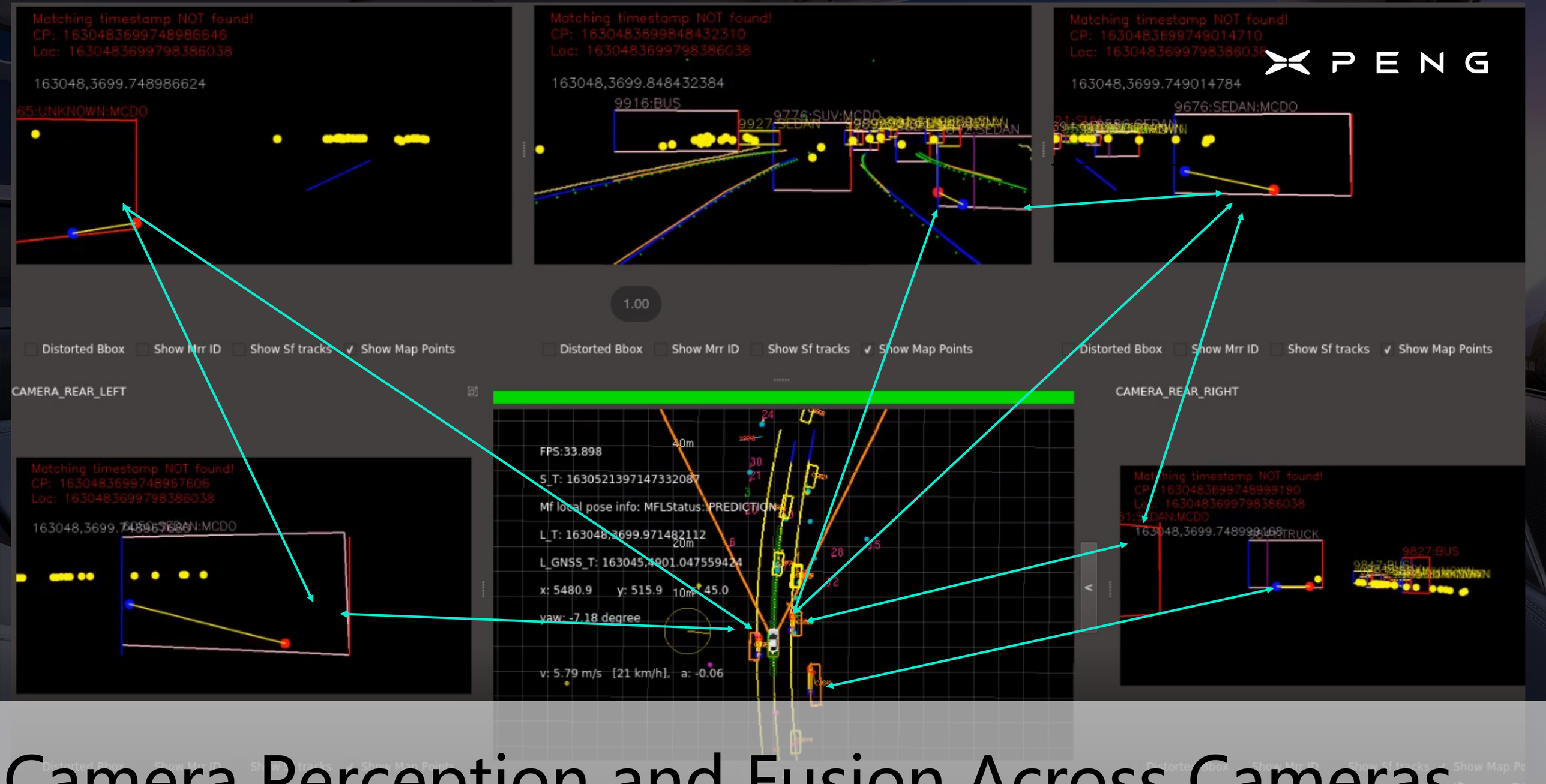
IMU, GPS, Wheel Odo

Ultrasound (12)

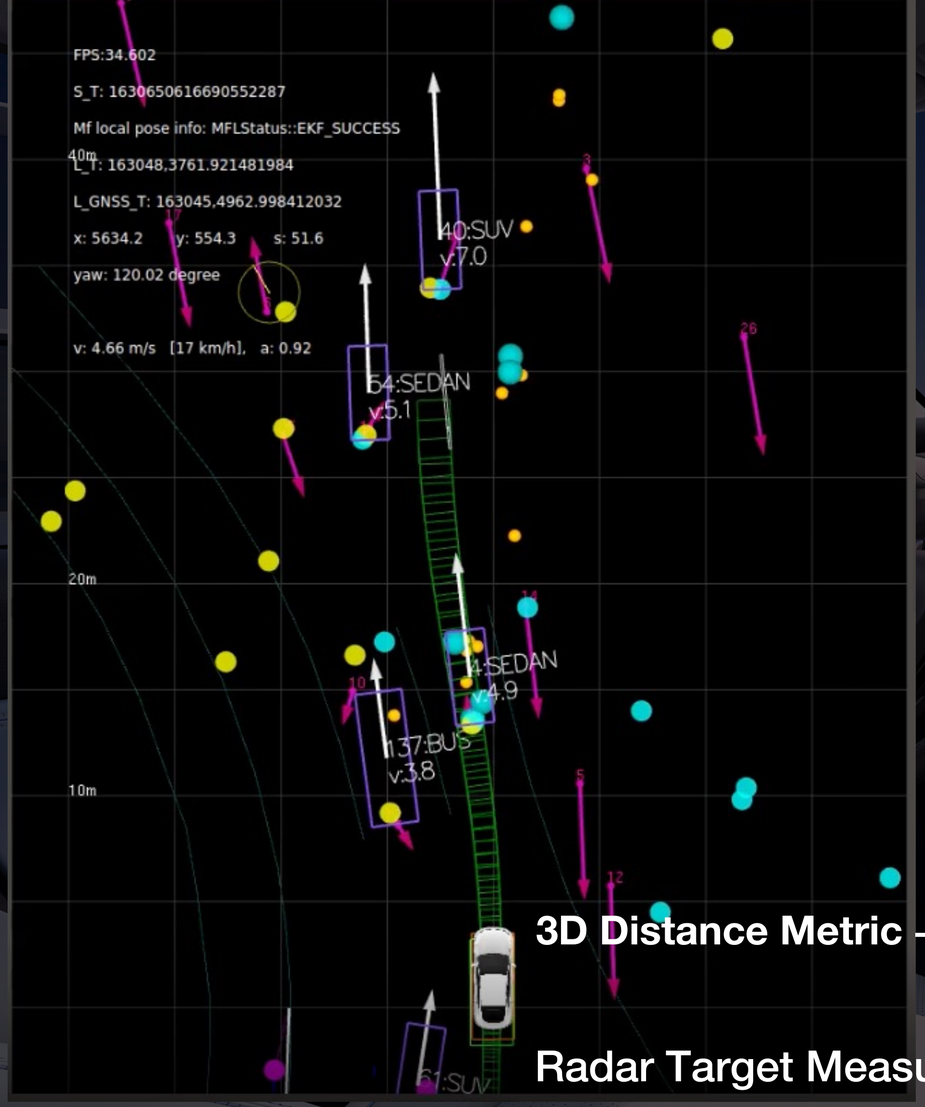
Actuators (lateral / long)

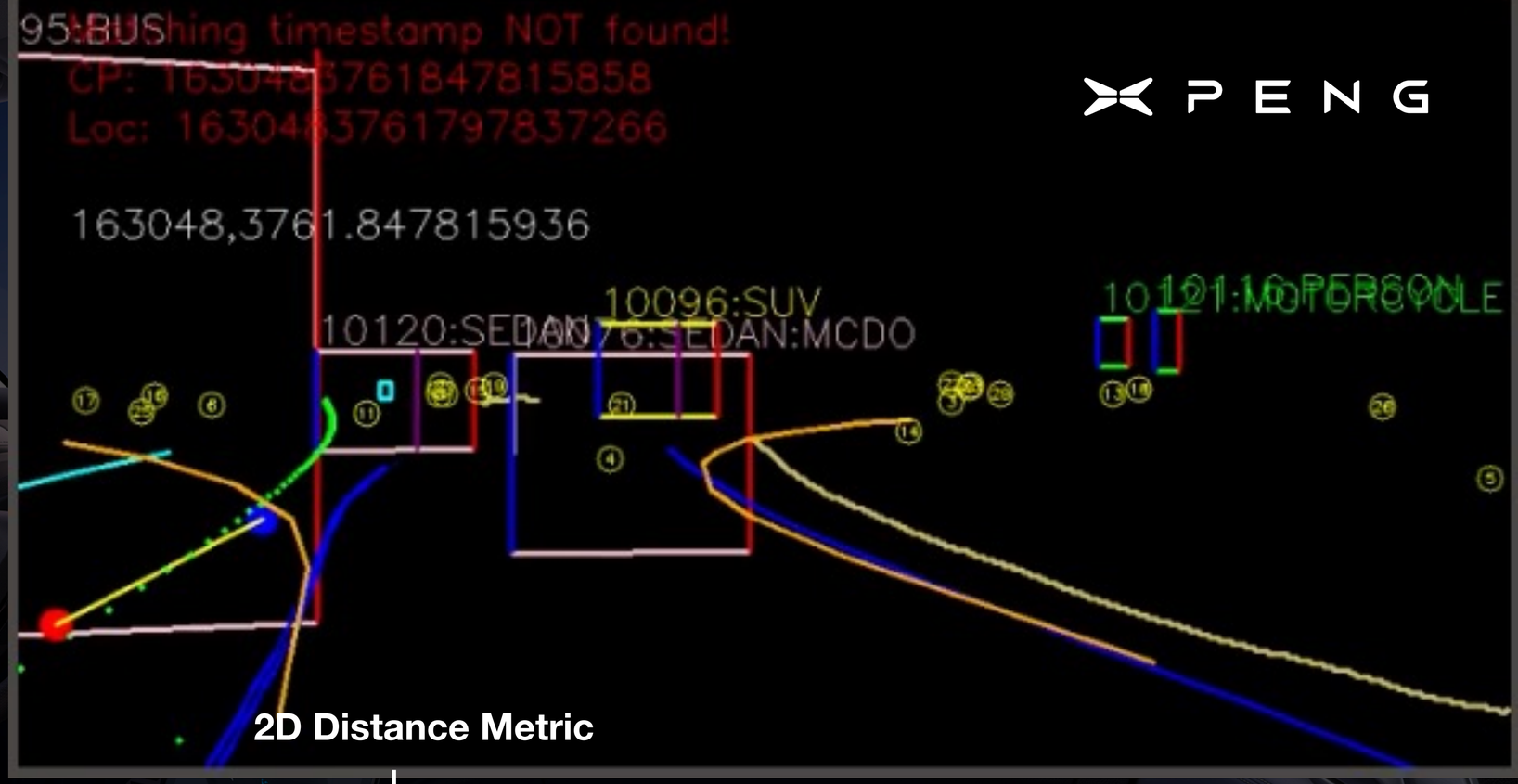
Main Sensors for CNGP





Camera Perception and Fusion Across Cameras





2D Distance Metric

3D Distance Metric → <Camera, Radar> Association → Targets

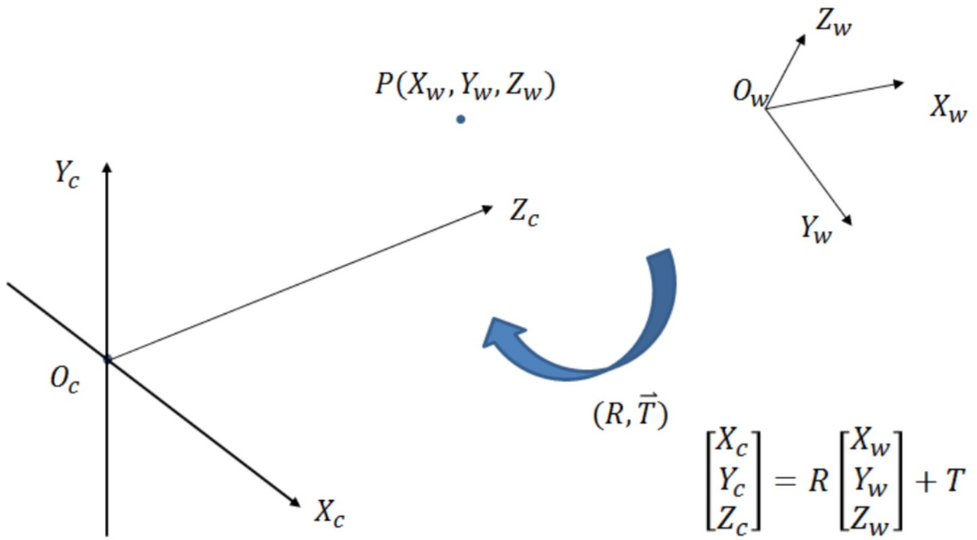
Radar Target Measurement: <rel_dist_X, rel_dist_Y, rel_Vx, rel_Vy, rel_ax, rel_ay, obstacle probability>

Radar Detection and Fusion with Camera



Lidar Perception and Fusion with Camera

Transform and Motion Compensation



Motion Compensation in Radar Measurement

Radar produces Relative Measurement in Radar reference frame



$$X_R = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \qquad V_R = \dot{X}_R = \begin{bmatrix} V_x \\ V_y \\ V_z \end{bmatrix}$$

How to recover target's Position and Linear Velocity in World Frame?

Position $X_w = RX_R + T$

- Radar frame's orientation in W frame
- Radar frame's translation in W frame

Localization Engine

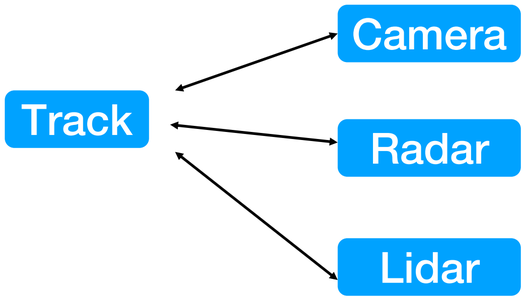
- Radar frame's orientation in W frame
- Radar frame's angular velocity in W frame
- Radar frame's linear velocity in W frame

Linear Velocity $V_w = RV_R + \dot{R}X_R + V_{R_in_W}$

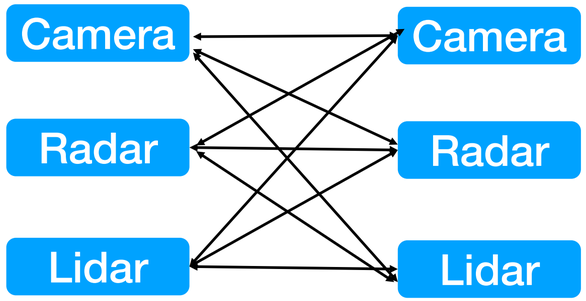
Data Association

Pairwise Distance Metric: dist<i, j>

Track to Measurement Association:

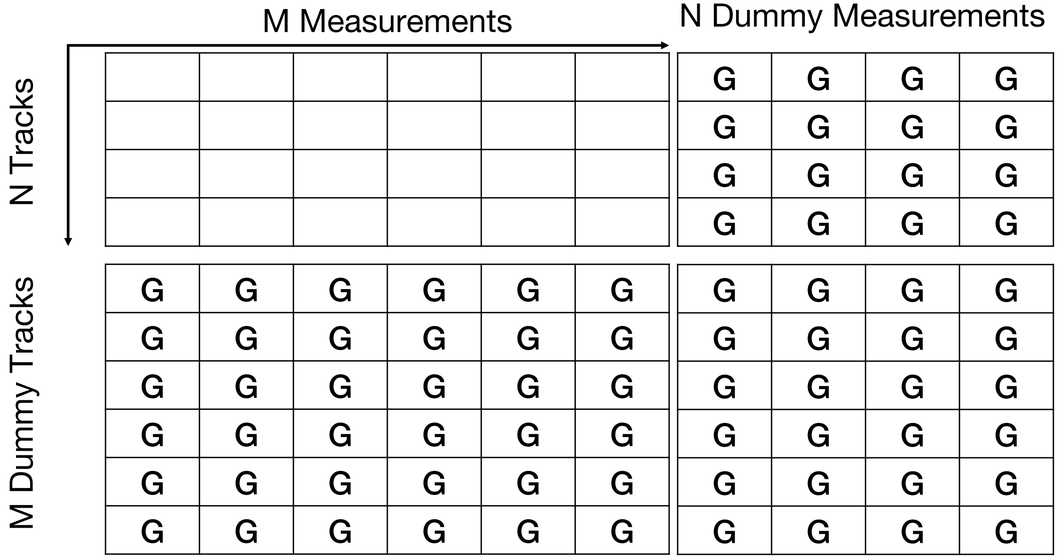


Measurement to Measurement Association:



Hungarian Method $O(n^3)$

Problem Statement: Given a nonnegative n-by-n cost matrix, where the <i,j>-th element represents the cost of assigning the j-th job to the i-th worker. Hungarian method aims to find an assignment such that each job is assigned to one worker, and each worker is assigned one job, such that the total cost of the assignment is minimized.



The original cost matrix can be augmented to avoid forcing association to happen with too large of a distance.

Extended Kalman Filter—Fusion Engine

Notation $\hat{\mathbf{x}}_{n|m}$ represents the estimate of $\mathbf{x}$ at time n given observations up to and including at time $m \le n$.

Predict

Predicted state estimate	$\hat{\boldsymbol{x}}_{k\|k-1} = f(\hat{\boldsymbol{x}}_{k-1\|k-1}, \boldsymbol{u}_k)$
Predicted covariance estimate	$\boldsymbol{P}_{k\|k-1} = \boldsymbol{F}_k \boldsymbol{P}_{k-1\|k-1} \boldsymbol{F}_k^\top + \boldsymbol{Q}_k$

Update

Innovation or measurement residual	$\tilde{\boldsymbol{y}}_k = \boldsymbol{z}_k - h(\hat{\boldsymbol{x}}_{k\|k-1})$
Innovation (or residual) covariance	$\boldsymbol{S}_k = \boldsymbol{H}_k \boldsymbol{P}_{k\|k-1} \boldsymbol{H}_k^\top + \boldsymbol{R}_k$
Near-optimal Kalman gain	$\boldsymbol{K}_k = \boldsymbol{P}_{k\|k-1} \boldsymbol{H}_k^\top \boldsymbol{S}_k^{-1}$
Updated state estimate	$\hat{\boldsymbol{x}}_{k\|k} = \hat{\boldsymbol{x}}_{k\|k-1} + \boldsymbol{K}_k \tilde{\boldsymbol{y}}_k$
Updated covariance estimate	$\boldsymbol{P}_{k\|k} = (\boldsymbol{I} - \boldsymbol{K}_k \boldsymbol{H}_k) \boldsymbol{P}_{k\|k-1}$

where the state transition and observation matrices are defined to be the following Jacobians

$$\boldsymbol{F}_k = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_{k-1|k-1}, \boldsymbol{u}_k}$$

$$\boldsymbol{H}_k = \left.\frac{\partial h}{\partial \boldsymbol{x}}\right|_{\hat{\boldsymbol{x}}_{k|k-1}}$$

Consider a Gaussian r.v. X, with $(\mu_x \quad \sigma_X^2)$, which we would like to estimate through the following observation model:

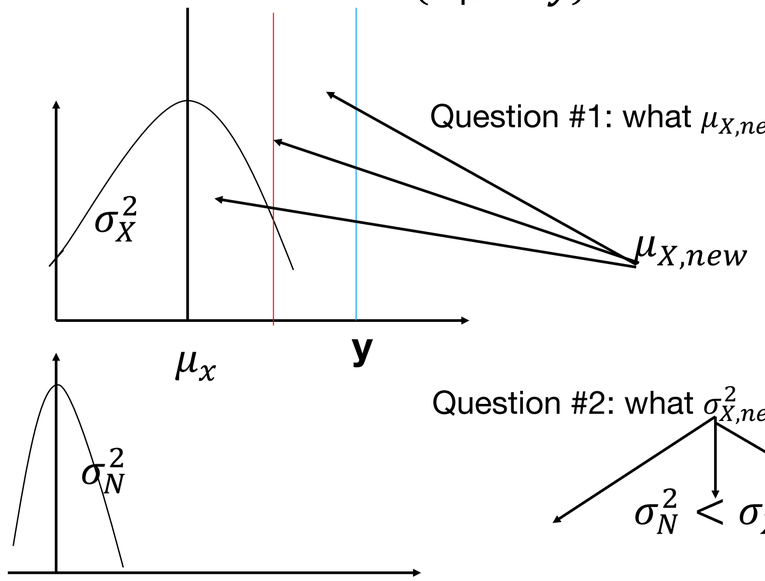
$$\boldsymbol{Y} = \boldsymbol{X} + \boldsymbol{N}$$

Where N is a AWGN with zero mean and variance of σ_N^2.

Think of Y as a sensor output, which produces a measurement of y as a specific realization of r.v. Y.

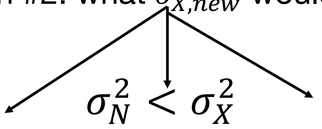
We assume that $\sigma_N^2 < \sigma_X^2$. After obtaining the measurement of y, the updated probabilistic belief will be updated as the following, which is still a Gaussian distribution. We would like to evaluate its updated mean and variance

$$\boldsymbol{P}(\boldsymbol{X}|\boldsymbol{Y} = \boldsymbol{y})$$

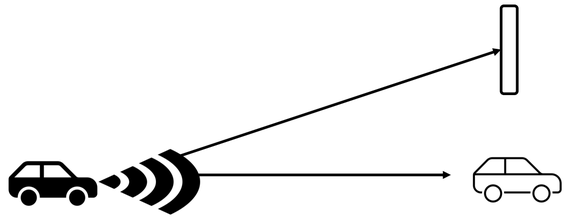


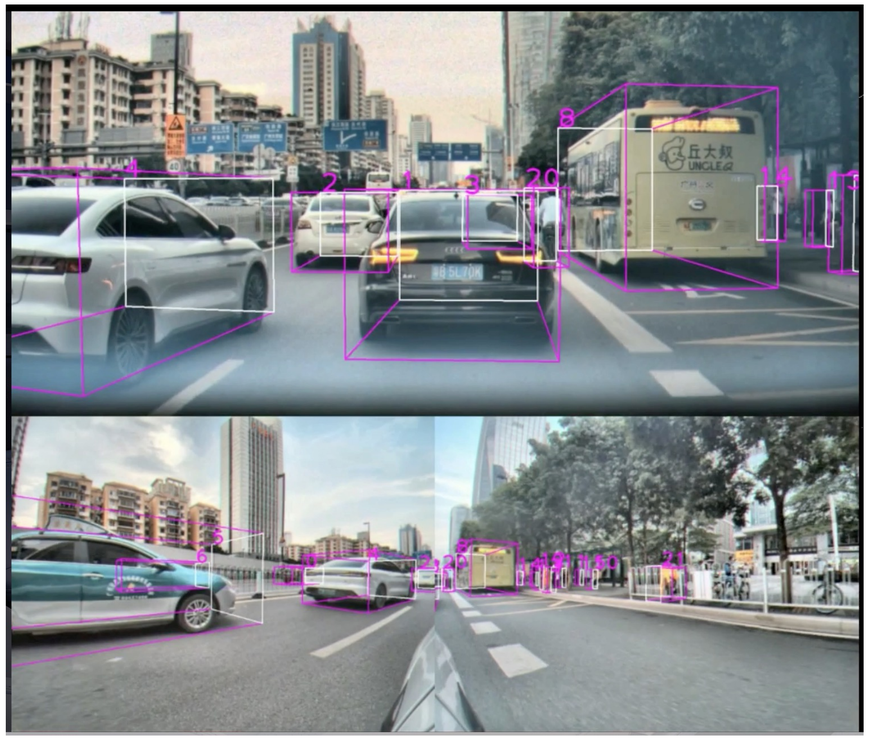
Question #1: what $\mu_{X,new}$ would look like?

Question #2: what $\sigma_{X,new}^2$ would look like?



Case Study: Stationary Object Detection





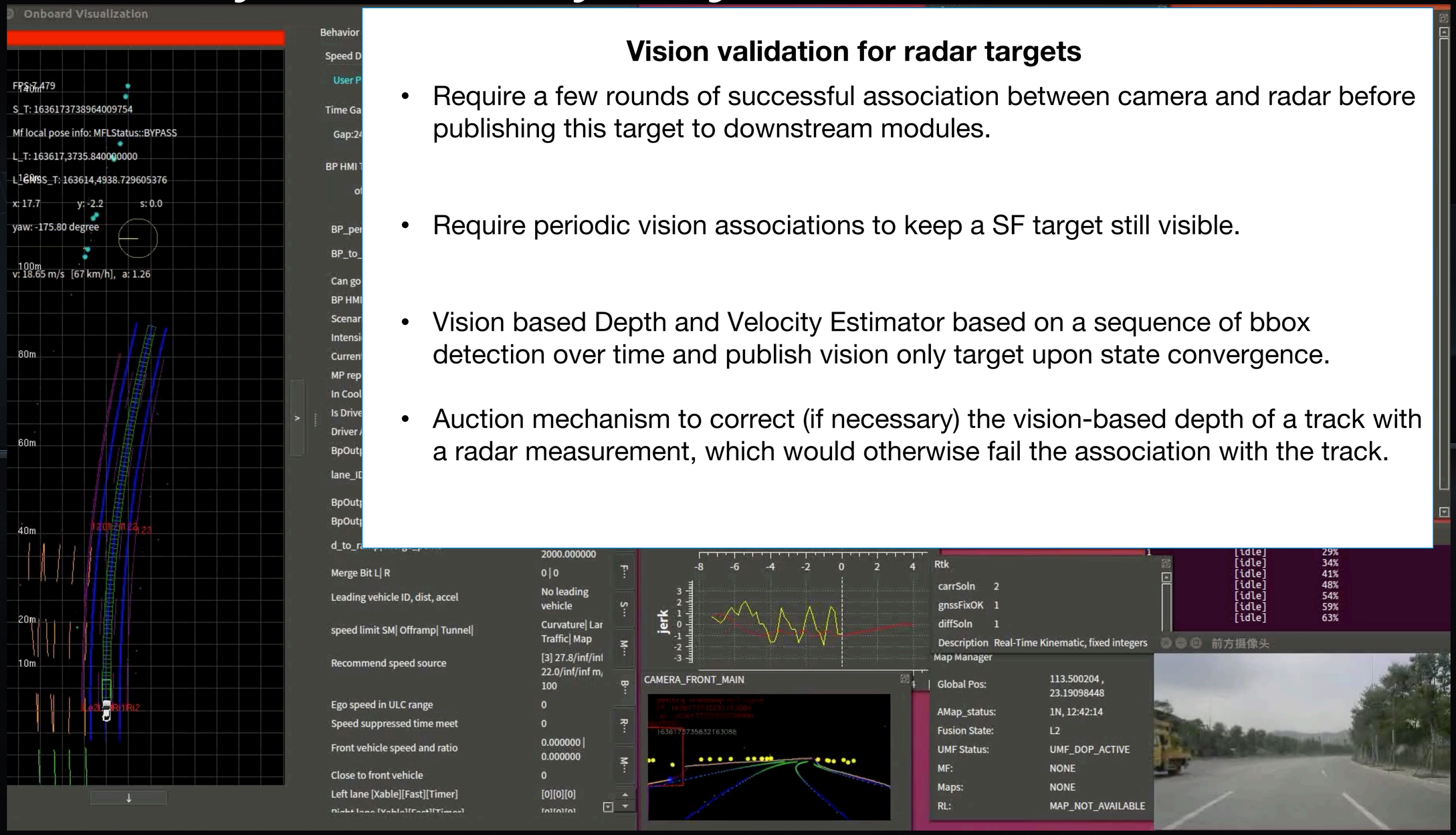
Why the stationary object detection is particularly challenging and important?

- Most of ADAS Radars do not have very good vertical resolution, if any.
- Radar targets can reflect from both stationary road users and static environment.
- Single-shot vision detection may see challenges with depth and scale ambiguity.
- Safety critical target requires reliable detection at 140+ meters at 120 KPH approaching speed for ego to react comfortably and confidently.

What we can do to solve it?

Explore the correlation across time and space (sensors)

Case Study: Stationary Object Detection



Vision validation for radar targets

- Require a few rounds of successful association between camera and radar before publishing this target to downstream modules.
- Require periodic vision associations to keep a SF target still visible.
- Vision based Depth and Velocity Estimator based on a sequence of bbox detection over time and publish vision only target upon state convergence.
- Auction mechanism to correct (if necessary) the vision-based depth of a track with a radar measurement, which would otherwise fail the association with the track.

Thank You

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